

Primary hypobetalipoproteinemia as a risk factor for liver complications

To the Editor:

Wargny *et al.* analysed extensive datasets from the French CONSTANCES database and the UK Biobank and concluded that primary hypobetalipoproteinaemia (HBL) is a risk factor for hard liver-related clinical outcomes.¹ While recognizing the value of this study we would like to make the following comments.

Firstly, although the Cox regression model is widely acknowledged for its utility, it may inadvertently lead to risk overestimation in scenarios beset with potential competing risks. Consequently, for the liver disease risk assessment explored in this paper, especially when liver-related diseases (such as cirrhosis and primary liver cancer) are potentially interrelated, employing a competing risks model appears more fitting. Traditional survival analysis techniques might not adequately account for the influence of secondary events on the primary study outcome, whereas the competing risks model affords a more holistic analytical viewpoint.^{2,3}

Secondly, the authors' inclusion of important information such as demographic characteristics, clinical examination findings, and biochemical data to adjust for potential covariates is commendable. However, we suggest a further expansion of the scope of covariates. Considering studies have shown differences in the treatment of cirrhosis among different races,⁴ it can be inferred that race might be an important covariate affecting the relationship between HBL and cirrhosis. Additionally, factors such as educational level⁵ and physical

activity⁵ should also be considered for a more comprehensive assessment of the stability and reliability of the results.

Third, this article primarily explores the relationship between HBL and liver disease complications in the general population.¹ Considering previous studies on type 2 diabetes and liver disease,⁶ conducting subgroup analyses might enhance the applicability and universality of these findings. The results indicated that low-density lipoprotein cholesterol levels increase with age, a trend particularly pronounced in women over 50¹. A detailed stratified analysis of this demographic could significantly benefit clinical practice. Furthermore, recommending stratified analyses for patients with hypertension and across various ethnic groups could deepen the understanding and evaluate the impact of these factors on research outcomes.

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Conflict of interest

The authors of this study declare that they do not have any conflict of interest.

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Authors' contributions

Mo-Yao Tan: conceptualization and manuscript draft. Xiang-Long Zhai: critical revision for important intellectual content, final approval.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.02.034>.

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